UNIVERSITAT POLITÈCNICA DE VALÈNCIA

Aprendizaje Automático Probabilístico

Probabilidad: modelos multivariados (actividades)

Alfons Juan

DSIC

Departamento de Sistemas Informáticos y Computación

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3.1. Distribuciones conjuntas para v.a. múltiples

3.1.1. Covarianza

- Halla la covarianza entre las variables aleatorias X e Y:

$p(X,Y)$	$Y=1$	$Y=2$	$Y=3$	$p(X)$
$X=1$	0.25	0.25	0	0.5
$X=2$	0	0.25	0.25	0.5
$p(Y)$	0.25	0.5	0.25	1.0

- Prueba que:

$$\mathbb{E}[\boldsymbol{x}\boldsymbol{x}^t] = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^t$$

3.1.2. Correlación

- Halla la correlación entre las variables aleatorias X e Y:

$p(X,Y)$	$Y=1$	$Y=2$	$Y=3$	$p(X)$
$X=1$	0.25	0.25	0	0.5
$X=2$	0	0.25	0.25	0.5
$p(Y)$	0.25	0.5	0.25	1.0

3.1.3. Correlación nula no implica independencia

- Prueba que, si dos variables aleatorias X e Y son independientes, entonces $\mathrm{Cov}[X, Y] = \mathrm{corr}[X, Y] = 0$

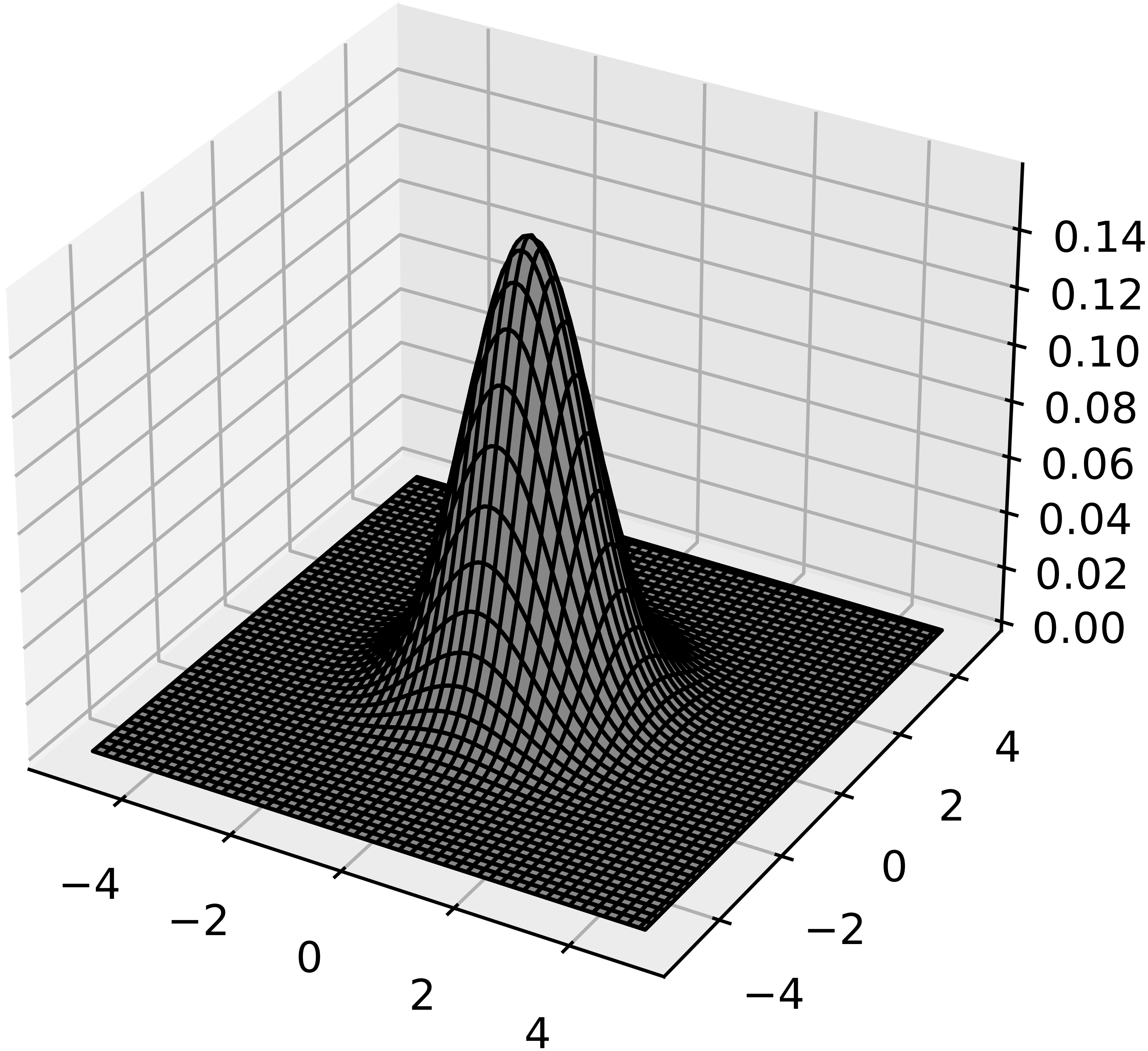
- Sean $X \sim U(-1,1)$ e $Y = X^2$, obviamente dependiente de X. Prueba que $\mathrm{Cov}[X,Y] = \mathrm{corr}[X,Y] = 0$. Pista: $\mathbb{E}[X^3] = 0$.

3.2. La Gaussiana (normal) multivariada

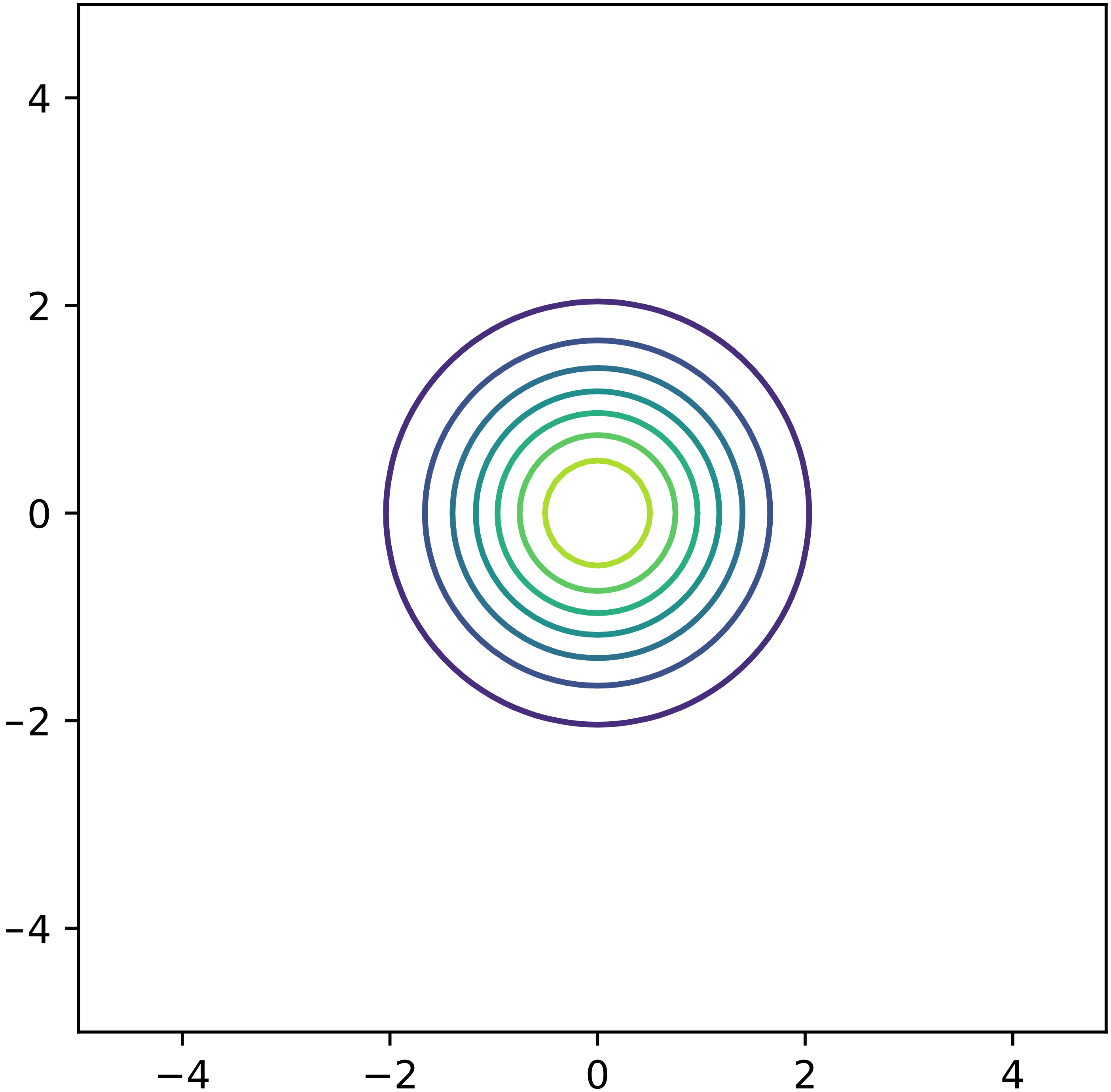
- *Σ esférica:* $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$, $\sigma^2 = 1$

3.2.N.py

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import multivariate_normal

m, S = [0, 0], [[1, 0], [0, 1]]
x, y = np.mgrid[-5:5:.1, -5:5:.1]
z = multivariate_normal(m, S).pdf(np.dstack((x, y)))

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(x, y, z, color='white',
 ↪  edgecolor="black")
plt.show()

fig = plt.figure()
ax = fig.add_subplot(111,aspect='equal')
ax.contour(x, y, z)
plt.show()
```

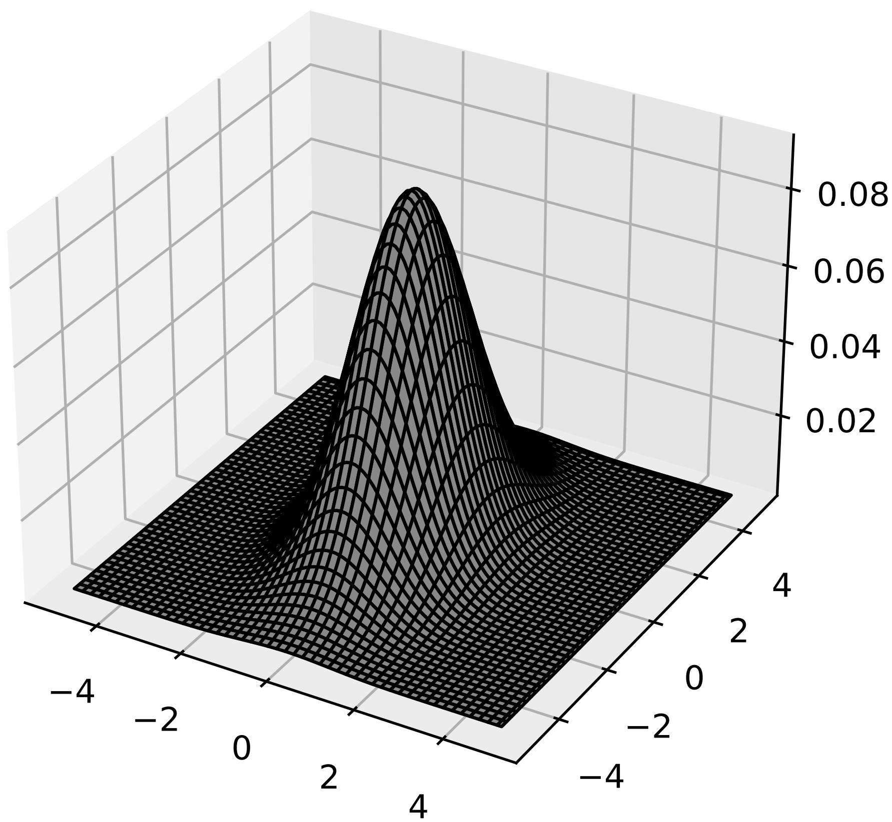


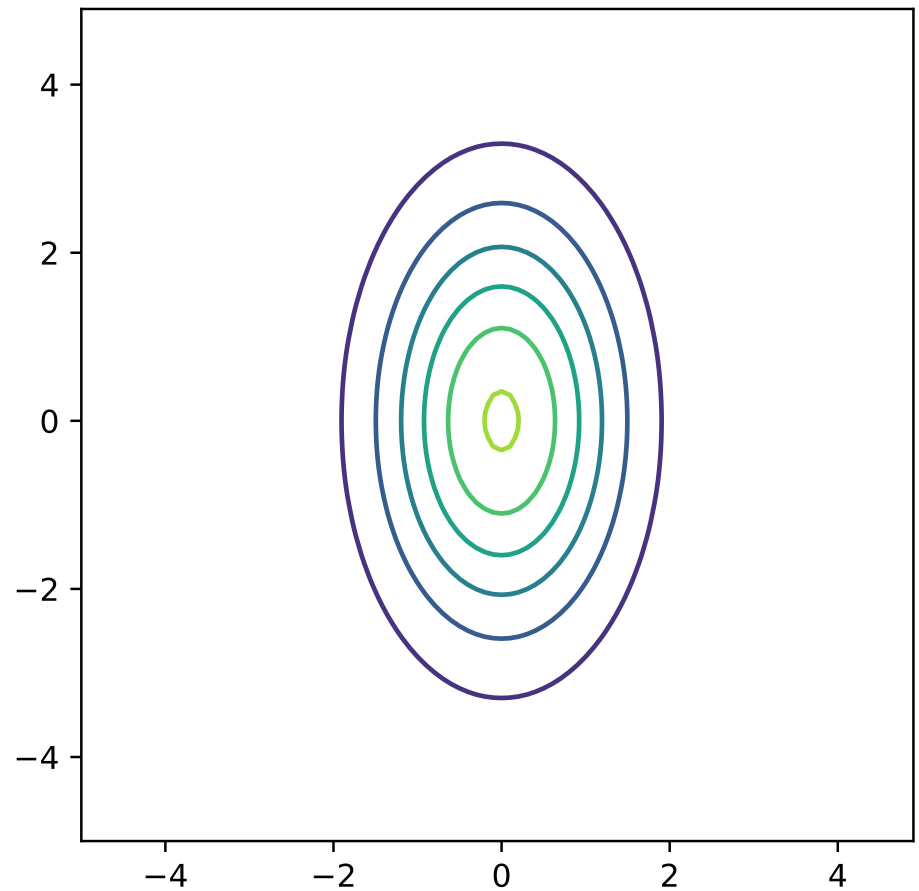
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−4
−2
0
2
4
4
2
0
−2
−4


4
2
0
−2
−4
−4
−2
0
2
4

- Modifica el script anterior para el caso Σ ***diagonal:***

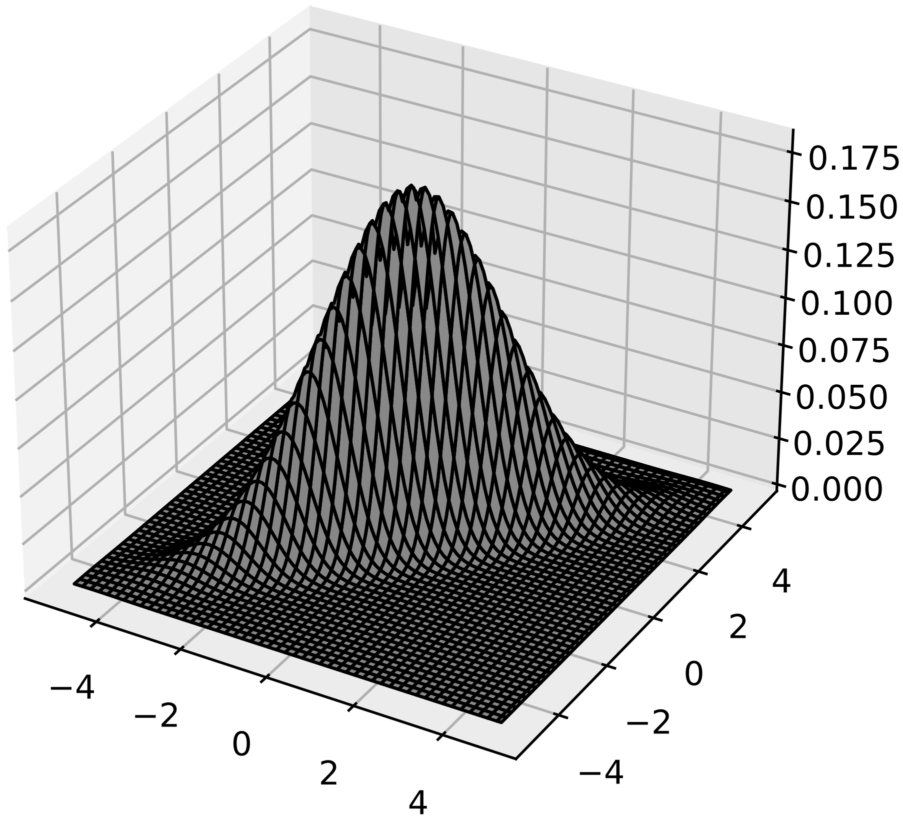
$$\boldsymbol{\mu} = \mathbf{0} \qquad \text{y} \qquad \boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$$

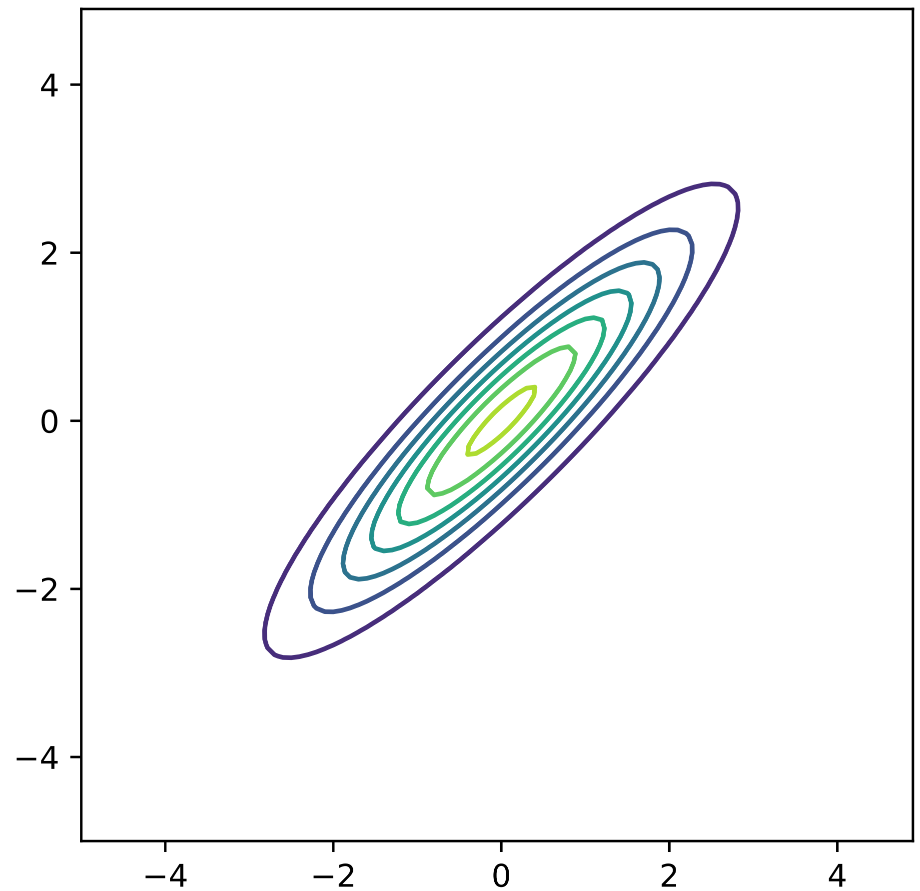




- Modifica el script anterior para el caso Σ ***completa:***

$$\boldsymbol{\mu} = \mathbf{0} \qquad \text{y} \qquad \boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1.8 \\ 1.8 & 2 \end{pmatrix}$$





▸ Descomposición en valores propios (espectral) de $\mathbf{\Sigma}$: $\mathbf{\Sigma} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^t$

3.2.eigh.py

```python
import numpy as np
S = np.array([[2, 1.8], [1.8, 2]])
La, U = np.linalg.eigh(S)
i = La.argsort()[::-1]; La = La[i]; U = U[:,i]
print(La, U, U @ np.diag(La) @ U.T)
```

3.2.eigh.out

```
[3.8 0.2] [[ 0.70710678 -0.70710678]
 [ 0.70710678  0.70710678]] [[2.  1.8]
 [1.8 2. ]]
```

▹ Comprueba que $\begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 3.8 & 0 \\ 0 & 0.2 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} 2 & 1.8 \\ 1.8 & 2 \end{bmatrix}$

- Bola de Mahalanobis centrada en $\boldsymbol{\mu}$, de $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^t$ y radio r:

$$\{\boldsymbol{y} : \mathcal{B}_{\Sigma^{-1}}(\boldsymbol{y}, \boldsymbol{\mu}) = r\} \qquad \text{(frontera de la bola)}$$

$$= \{\boldsymbol{y} : (\boldsymbol{y} - \boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1} (\boldsymbol{y} - \boldsymbol{\mu}) = r\}$$

$$= \{\boldsymbol{y} : \boldsymbol{z} = \boldsymbol{y} - \boldsymbol{\mu},\ \boldsymbol{z}^t \boldsymbol{\Sigma}^{-1} \boldsymbol{z} = r\}$$

$$= \{\boldsymbol{y} : \boldsymbol{z} = \boldsymbol{y} - \boldsymbol{\mu},\ \boldsymbol{z}^t \mathbf{U}\boldsymbol{\Lambda}^{-1}\mathbf{U}^t \boldsymbol{z} = r\}$$

$$= \{\boldsymbol{y} : \boldsymbol{z} = \boldsymbol{y} - \boldsymbol{\mu},\ \underbrace{\boldsymbol{z}^t \mathbf{U}\boldsymbol{\Lambda}^{-1/2}}_{\boldsymbol{x}^t} \underbrace{\boldsymbol{\Lambda}^{-1/2}\mathbf{U}^t \boldsymbol{z}}_{\boldsymbol{x}} = r\}$$

$$= \{\boldsymbol{y} : \boldsymbol{x} = \boldsymbol{\Lambda}^{-1/2}\mathbf{U}^t(\boldsymbol{y} - \boldsymbol{\mu}),\ \boldsymbol{x}^t \boldsymbol{x} = r\}$$

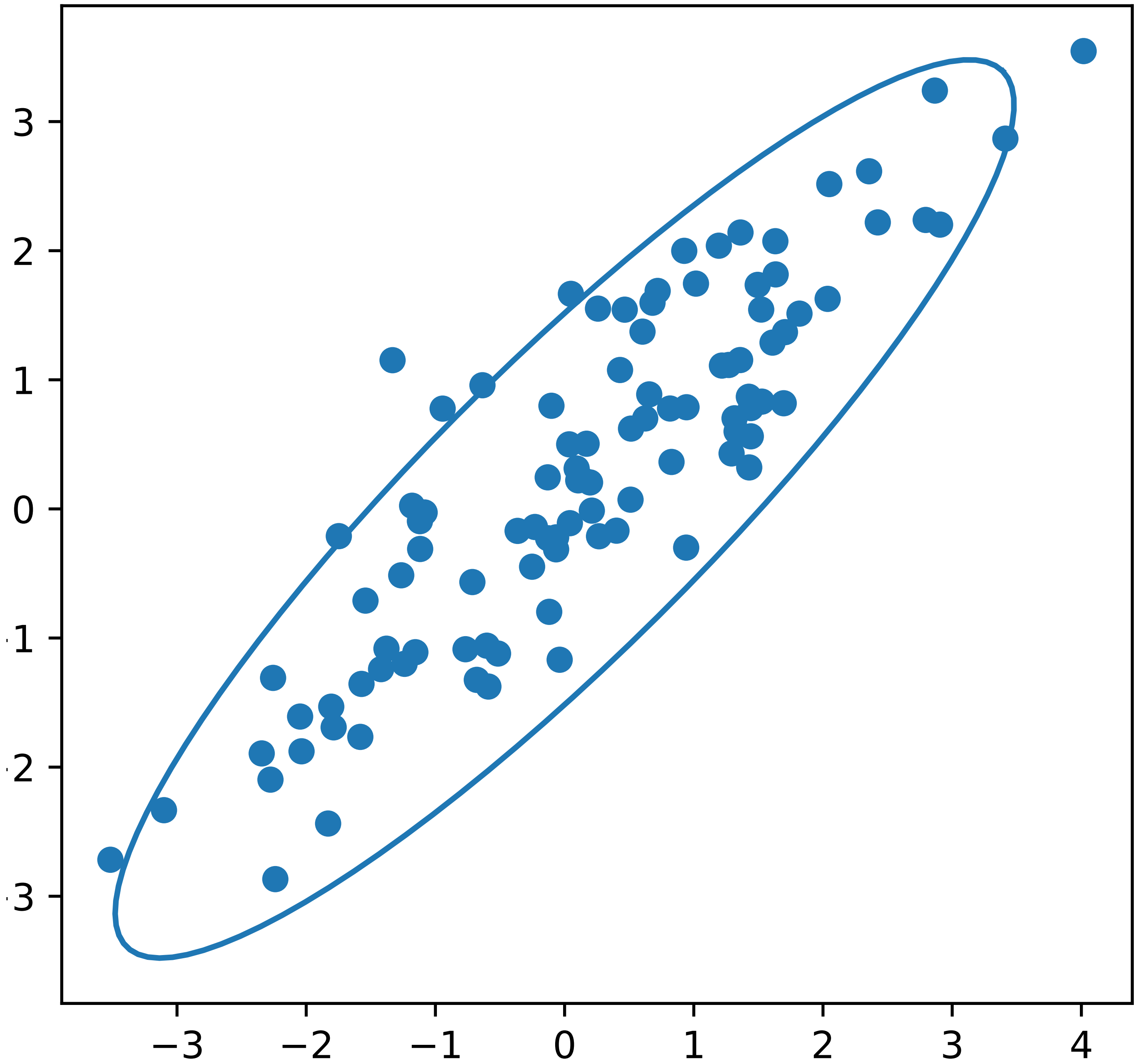
$$\overset{2D}{=} \left\{\boldsymbol{y} : \boldsymbol{\Lambda}^{-1/2}\mathbf{U}^t(\boldsymbol{y} - \boldsymbol{\mu}) = \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix}, 0 \leq \theta \leq 2\pi\right\}$$

$$= \left\{\boldsymbol{y} : \boldsymbol{y} = \boldsymbol{\mu} + \mathbf{U}\boldsymbol{\Lambda}^{1/2} \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix}, 0 \leq \theta \leq 2\pi\right\}$$

$\triangleright$ $\boldsymbol{\mu} = \mathbf{0}$, $\mathbf{U} = \begin{pmatrix} .71 & -.71 \\ .71 & .71 \end{pmatrix}$, $\boldsymbol{\Lambda} = \operatorname{diag}(3.8, .2)$ y $r = 2.45$ (95 %):

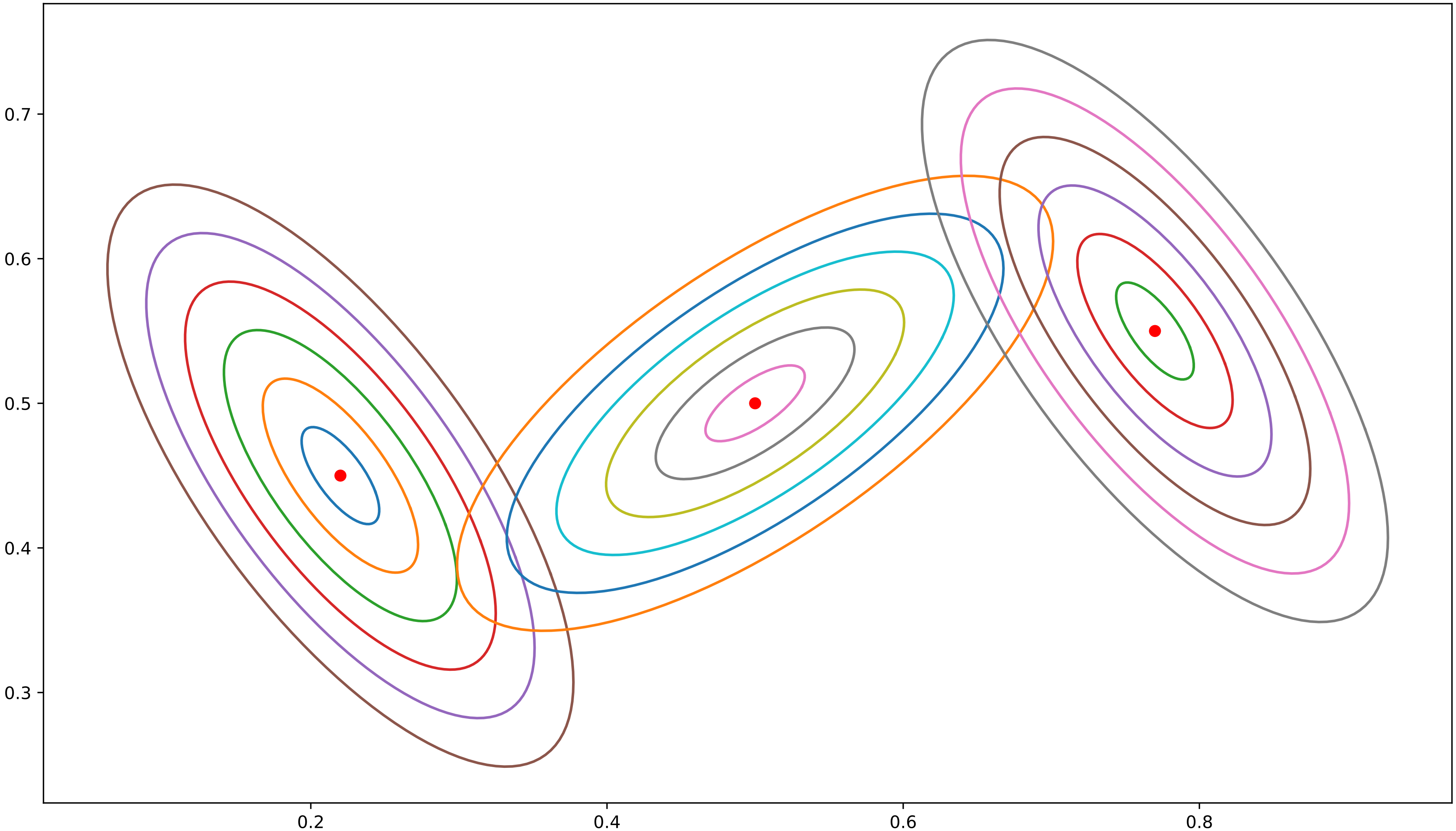
3.2.Mahalanobis.py

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import multivariate_normal

U = np.array([[.71, -.71], [.71, .71]])
La = np.array([3.8, .2])
r = 2.45
t = np.linspace(0, 2*np.pi, 100)
x, y = U @ np.diag(np.sqrt(La)) @
  np.array([r*np.cos(t), r*np.sin(t)])
Y = np.vstack((x, y)).T
cov = U @ np.diag(La) @ U.T
X = multivariate_normal(cov=cov).rvs(100)
fig = plt.figure()
ax = fig.add_subplot(111, aspect='equal')
plt.plot(Y[:, 0], Y[:, 1])
plt.scatter(X[:,0],X[:,1])
plt.show()
```


3
2
1
0
-1
-2
-3
−3
−2
−1
0
1
2
3
4

3.5. Modelos de mixtura

- Reproduce el cuaderno de la Figura 3.11



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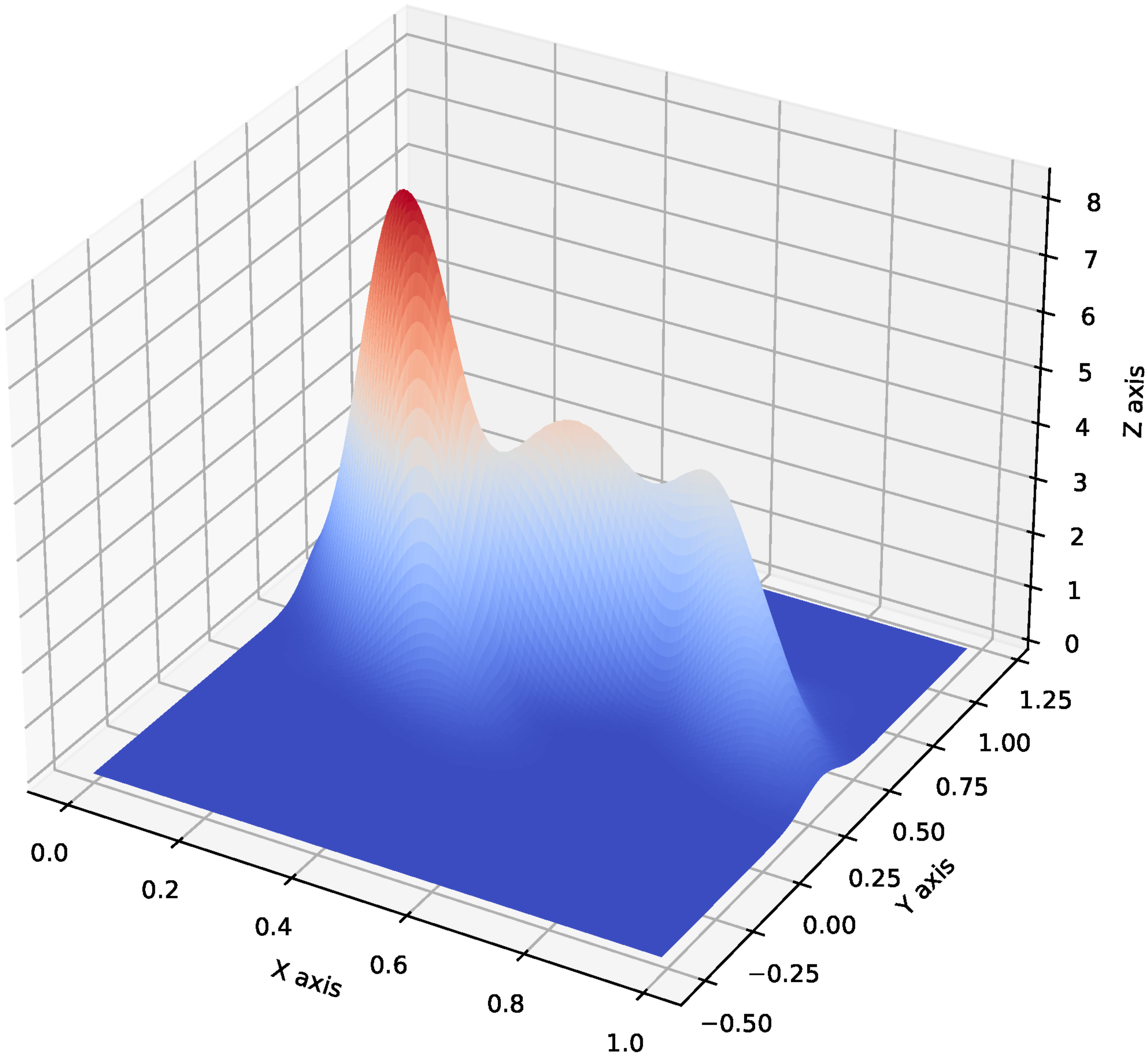
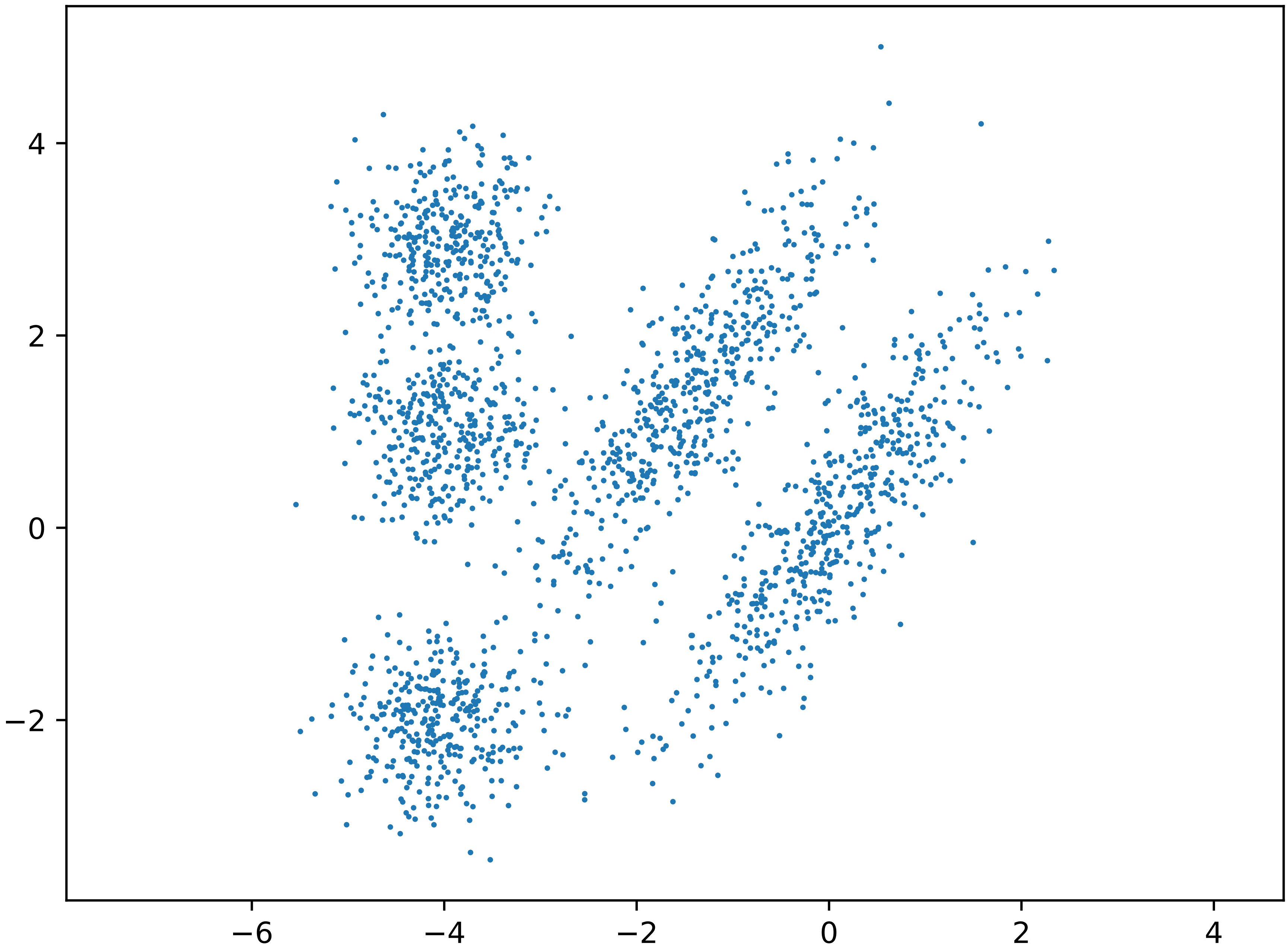
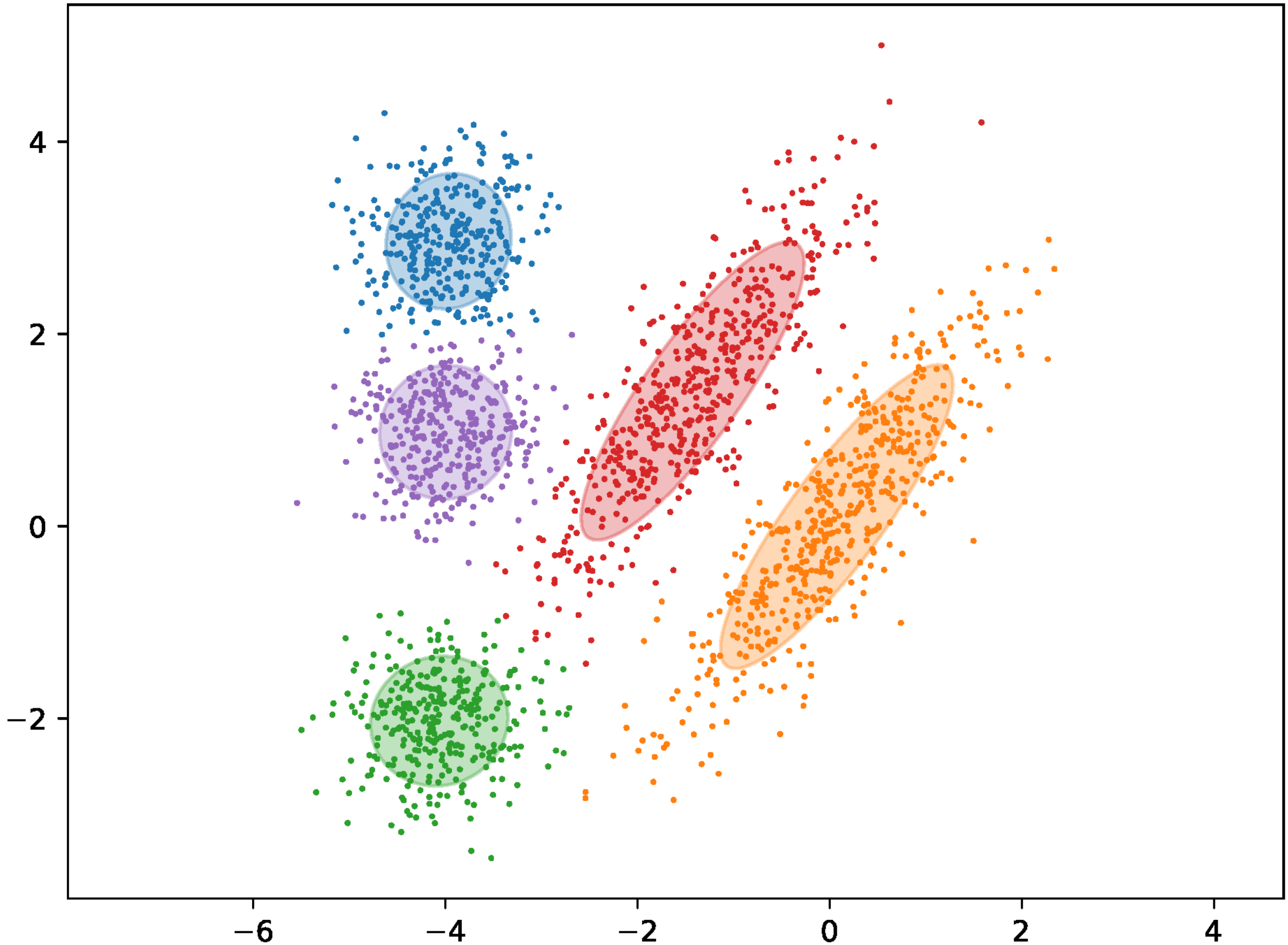


8
7
6
5
4
3
2
1
0
Z axis
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
Y axis
0.0
0.2
0.4
0.6
0.8
1.0
X axis

- Reproduce el cuaderno de la Figura 3.12




4
2
0
−2
−6
−4
−2
0
2
4

- Reproduce el cuaderno de la Figura 3.13